DSRNet: Diagonal Subsampling Reconstruction Network for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection seeks to locate pixels in a scene exhibiting substantial spectral discrepancies from the surrounding background pixels, holding essential applications in both civilian and military domains. However, in real-world scenarios, the inherent properties of hyperspectral imaging, the irregular forms of anomalous targets, and the lack of prior information pose significant challenges for anomaly detection methodologies. To address these issues, we first explore a novel reconstruction-based modeling approach for hyperspectral anomaly detection, offering a rational motivation for the modeling approach and a detailed exposition of its effective implementation. Furthermore, we propose a diagonal subsampling reconstruction network (DSRNet) for anomaly detection of hyperspectral data. Specifically, DSRNet consists of a paired training data generation algorithm using subsampling and a self-supervision training process enforced with a reconstruction consistency constraint. The training input is derived by a randomly diagonal averaging subsampler, where training pairs are derived from the same original hyperspectral data. Extensive experiments on four public datasets demonstrate the superiority of our DSRNet compared over several state-of-the-art baselines, with an average AUC score increase of 0.0156.**

***Index Terms*— Anomaly detection, hyperspectral image (HSI), image subsampling, neural network, self-supervision learning.**

I. Introduction

REMOTE sensing serves as a situational awareness technology, which operates at multiple levels from multiple perspectives, playing a crucial role in collecting environmental information about the earth [1]. Hyperspectral remote sensing, positioned as an advanced technology in the field, has surpassed the constraints of traditional visible and infrared sensors [2], [3], [4]. It spans numerous spectral bands, encompassing the imperceptible segment of the electromagnetic spectrum, allowing for a more consistent alignment with the physical characteristics of ground mapping information [5], [6]. Hyperspectral anomaly detection is capable of identifying anomalous pixels within images through the analysis of spectral discrepancies, without relying on any prior information [7]. Consequently, it has emerged as a significant study focus in the domain of intelligent perception within remote sensing data, offering unique insights for the precise detection and perception of targets [8], [9]. With a rising requirement of reconnaissance technologies, the complexity of scenes and targets in hyperspectral images (HSIs) is also increasing. This complexity imposes higher demands on anomaly detection algorithms, thus positioning them as a focal point of study and a formidable challenge.

A considerable portion of ongoing efforts is dedicated to the advancement of robust and efficient anomaly detection technologies. This encompasses approaches grounded in statistical methods, representation models, and deep neural networks. Statistical models were widely applied to extract anomaly targets in the early studies. The Reed-Xiaoli (RX) [10] defined a hypothesis which the background obeys a multivariate Gaussian distribution, whereas the anomalies deviate from this distribution. Nevertheless, the intricate and dynamic backgrounds pose a challenge in precisely characterizing their distribution features using the RX algorithm. To further enhance the representation ability of the model in complex scenarios, a series of improved versions have optimized the RX algorithm from multiple perspectives [11], [12], [13], [14], [15], [16], [17]. The segmented RX [11] method leverages the adjacent pixel within the clustering algorithm to detect anomalies. The local RX (LRX) [12] constructs statistics exclusively from the adjacent pixels of the validation data, rather than considering pixels in the entire scene. Tao et al. [17] carried out a transition of all pixels to a temporary state, such as the fractional Fourier domain, prior for executing detection. Yang et al. [18] integrates the detection outcomes of various characteristics using the RX algorithm with the idea of ensemble to extract the complete characteristics. While statistical model-based anomaly detection methods have demonstrated encouraging result, they still fall short in accurately representing the intricate distribution of hyperspectral data.

On the other hand, representation learning shows enhanced performance in hyperspectral anomaly detection, including collaborative representation [19], [20], [21], sparse representation [22], [23], [24], [25], and low-rank representation [26], [27], [28], [29], [30], [31]. Li and Du [32] presented the collaborative representation detector (CRD), which employs a similarity matrix to optimize the representation and detect anomalous. Given anomalous instances are susceptible to interference from background information generated by CRD, Zhao et al. [33] transformed the source target to the fractional Fourier domain and devised an innovative weighted matrix to accommodate the priorities of backgrounds. Hou et al. [34] proposed a background purification strategy (CRDBPSW)

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aimed at mitigating the influence of potential aberration pixels, thus creating an effective enhanced version based on CRD. Gao et al. [35] accomplish precise anomaly detection by autonomously devising a topological approach shaped like a chessboard that reduces the reliance on hypothesizes and immediately extracts characteristics from original data. They successfully addresses the shortcomings of conventional methods, making a substantial progress in anomaly detection domain. Yuan et al. [36] introduced anomaly method leveraging the sparse concept, which designs a double window architecture and identifies anomalous targets by computing its sparse factor. However, algorithms developed based on sparse theory cannot achieve the optimal output results. In response to this limitation, Niu and Wang [37] employed the joint use of low-rank representation and dictionary learning to extract features in hyperspectral data. Ma et al. [38] presented a detection algorithm that integrates fractional Fourier shift with low-rank representation. The appearance of low-rank representation in anomaly detection has given rise to numerous associated algorithms [39], [40], [41], [42]. Guo et al. [43] enhanced anomaly detection by employing structural incoherence and mixed noise models to separate background and anomaly pixels. The growing demand of applications contributes to an progressively intricate background constraining the effectiveness of conventional algorithms.

Deep learning-based approaches stand out in anomaly detection because they are suitable to extract high-level features from both backgrounds and targets [44], [45]. Li et al. [46] introduced a strategy for anomaly detection utilizing a convolutional neural network (CNN) that trained using pairs of data derived from annotated instances. Song et al. [47] designed a innovative hyperspectral anomaly detection algorithm that relies on an enhanced low-rank representation and density features derived using a CNN framework. For the acquisition of a distinguishing characteristic, Xie et al. [48] proposed an adversarial autoencoder to highlight the features of anomalies. LREN [49] further embeds physical properties into the network architecture, building density estimation modules to better characterize the learned potential representations. Jiang et al. [50] constructed an unsupervision architecture using generative adversarial network to accomplish better generative results. One viable solution is an autoencoder-based approach that detects anomalies by processing raw data, where the discrepancy between the original input data and the generated data is treated as an anomaly degree. The robust graph autoencoder (RGAE) [51] was developed to bolster the robustness of networks against anomalous noise and targets through the incorporation of hypergraphs. Jiang et al. [52] presented an integrated architecture (E2E-LIADE) of end-to-end autoencoder density evaluation in the domain of anomaly detection. Xiao et al. [53] comprehensively captured characteristics from both global and local interpretations for anomaly detection. Gao et al. [35] introduced an innovative blind-spot framework designed to boost the semantic information extraction of targets while diminishing the characteristic extraction of the background. Wang et al. [54] developed an innovative subsampling strategy, which constructed a robust anomaly detection network. Xiang et al. [55] introduced an auxiliary component

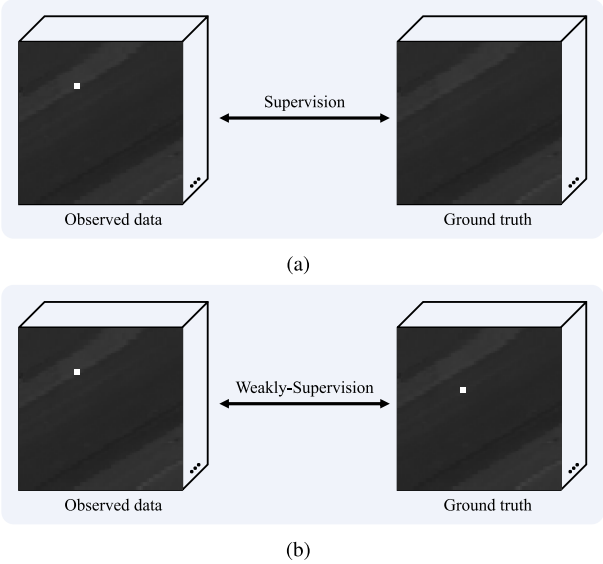


Fig. 1. Supervision approach and weakly supervision approach for hyperspectral anomaly detection based on background reconstruction. (a) Supervision. (b) Weakly supervision.

on the underlying characteristics in the autoencoder (GAED) to boost its capability to reconstruct hyperspectral data. Ma et al. [56] designed a characteristic dynamic construction component and a graphics-guided selection component to improve mutual information within original hyperspectral data.

Due to the inherent properties of hyperspectral imaging and the irregular forms of anomalous targets, employing deep learning architectures for anomaly detection remains a tough challenge. There are two challenges associated with the advancement of deep learning-based anomaly detection approaches.

1) Existing spectral reconstruction-based deep learning detection algorithms face limitations in constructing effective feature extractions, often capturing low-level features between the background and anomalies. Therefore, such a drawback may lead to learning an identity mapping between input and output, eventually failing to detect anomalous targets.
2) The high-dimensional nature of hyperspectral data presents a challenging tradeoff in practical applications. Due to the low efficiency of characteristic expressions, the existing anomaly detection models are becoming increasingly intricate. Such deficiency eventually results in a substantial amount of computing consumption, hindering real-time onboard processing, and severely restricting practical applications.

In brief, a lightweight and effective self-supervised solution is needed for hyperspectral anomaly detection algorithms suitable in practical applications. As shown in Fig. 1(a), for supervised learning methods, we require a hyperspectral background data of the completely identical scene as ground truth, or manually annotate regions as background in an HSI containing anomalous targets. Hence, theoretically, supervised methods can achieve state-of-the-art performance through training a network end-to-end to map an image containing

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anomalous targets to a clean background image. However, the supervision approach consumes substantial manual annotation efforts and experimental resources, limiting its widespread applicability in practical scenarios. Consider a slightly more relaxed setting. In the absence of any ground truth, the training process can be conducted on two HSIs containing abnormal targets in different positions of the same static scene, and the background can also be reconstructed. Training a reconstruction network to map one image with anomalous targets to another image with anomalous targets of the same scene will perform as well as mapping to the ground truth. See Fig. 1(b) for an illustration of the weakly supervision setting. The mathematical analysis and proof are provided in detail in Section II-C.

To tackle the above challenges, we propose a Diagonal Subsampling Reconstruction Network (DSRNet) for anomaly detection on hyperspectral data. Specifically, we leverage a novel background reconstruction-based technical framework, which comprises a paired training data generation algorithm using subsampling and a self-supervision training process enforced with a reconstruction consistency constraint. The model input is obtained by a randomly diagonal averaging subsampler, where training pairs are derived from the same hyperspectral data and the elements corresponding to the two instances are cross-adjacent in the original data. Furthermore, we formulate a reliable optimization constraint function for ensuring the hyperspectral reconstruction consistency by rigorous mathematical modeling and theoretical proof. In the detection scheme, the trained model is considered as a background reconstruction network, which tends to reconstruct pixels at anomalous target positions in the HSI as background pixels through the extracted global characteristics. Ultimately, the anomaly level of each position in hyperspectral data can be assessed using the overall reconstruction error of image pixels. To validate the effectiveness of the proposed DSRNet, we execute extensive experiments on four available benchmark datasets. The results indicate the superiority of our proposed DSRNet compared with several state-of-the-art approaches.

The main contributions of our article can be summarized as follows.

1) From the theoretical perspective, we explore a novel modeling approach for hyperspectral anomaly detection. Through rigorous mathematical derivations, we provide a rational motivation for the modeling approach and a detailed exposition of its effective implementation.
2) We propose DSRNet for hyperspectral anomaly detection, which comprises a paired training data generation algorithm using subsampler and a self-supervision training process enforced with a reconstruction consistency constraint. Our algorithm is implemented directly on the original data without relying on any prior knowledge.
3) Extensive experiments on four benchmark datasets indicate the superiority of our proposed algorithm compared with several state-of-the-art approaches. The concise architecture design and efficient anomaly detection performance can demonstrate its potential applications in real-world scenarios.

The remainder of this article is organized as follows. Section II presents the proposed modeling and framework in detail. Experiments are performed over four public hyperspectral data to verify the effectiveness of the proposed method in Section III. Finally, conclusion is presented in Section IV.

II. Methodology

In this section, we first present an overview of our approach. Subsequently, we provide detailed implementation specifics for each component.

A. Overview

The overview framework of DSRNet is illustrated in Fig. 2, which consists of three major components: the neural network architecture, the training scheme, and the detection scheme. For a given original unlabeled HSI with $R \times C$ pixels and B spectral bands, we denote it as $\mathbf{X} \in \mathbb{R}^{R\times C\times B}$. Our idea is to train a self-supervision background knowledge reconstruction model, which means no context and anomalous target label knowledge is needed, only the observed data $\mathbf{X}$ is adopted. The training input is derived by a randomly diagonal averaging subsampler, where training pairs are derived from the observed $\mathbf{X}$. Our network learns feature representations of the background, with high reconstruction error for anomalous targets and low reconstruction error for background pixels. Therefore, the reconstruction error can indicate the anomaly level of pixels and help identify anomalous targets.

B. Neural Network Architecture

Numerous self-supervision and upsupervision anomaly detection approaches adopt a relatively complex architecture, such as a generative adversarial network or encoder–decoder network. Conversely, we employ an extremely concise three-layer regular convolutional network. The neural network structure of the proposed method is illustrated in Fig. 3. The architecture comprises merely two 3×3 convolution, succeeded by one 1×1 convolution. The Leaky Rectified Linear Unit (LReLU) layer in sequence follows each convolutional layer. Beyond that, there are no normalization layers or attention modules. The architecture is characterized by approximately 10k parameters, a size considered clear in comparison to common anomaly detection models. The clear and parameter-less model facilitate quick detecting anomalous targets in real hyperspectral scene conducted on onboard devices.

C. Training Scheme and Analysis

Since the neural network is trained solely on original unlabeled HSI $\mathbf{X}$, we should obtain multiple paired instances from $\mathbf{X}$, where these paired instances are not the same as $\mathbf{X}$ and encapsulate the majority of its knowledge. To achieve this objective, we generate a set of diagonal subsampled image pairs of $\mathbf{X}$, denoted as $\{s_1(\mathbf{X}), s_2(\mathbf{X})\}$. The schematic illustration of paired instances using diagonal subsamplers is provided in Fig. 4.

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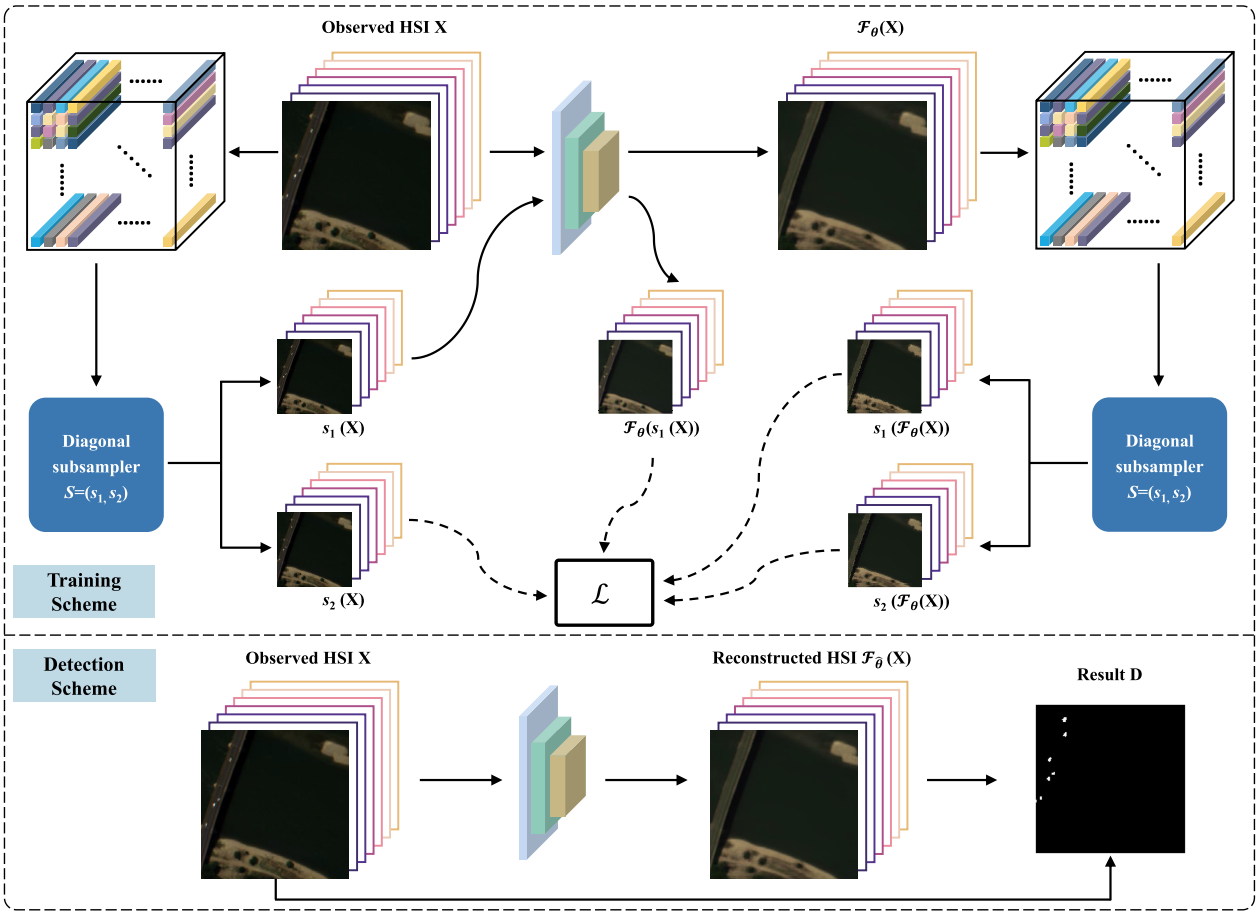


Fig. 2. Framework of the proposed DSRNet for hyperspectral anomaly detection. It consists of the training scheme and detection scheme. In the training scheme, the algorithm encompasses a paired training data generation algorithm based on subsampling and a self-supervised training approach enforced with a reconstruction consistency constraint. The training input is derived by a randomly diagonal averaging subsampler. In the detection scheme, the trained model tends to reconstruct pixels at anomalous target positions as background pixels. The anomaly level of each position can be assessed using the overall reconstruction error of image pixels.

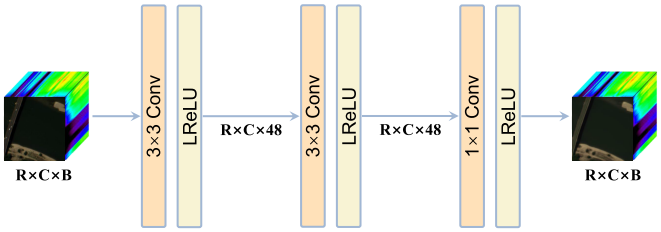


Fig. 3. Details of the proposed overall network architecture $\mathcal{F}_\theta$.

Represent an original hyperspectral as $\mathbf{X} \in \mathbb{R}^{R\times C\times B}$ with height R, width C, and total band B. The subsampler constructs those subsampled instances by partitioning the hyperspectral data into disjoint cubes of shape $m \times m \times B$. For each cube, the mean of the diagonal elements is randomly assigned to one of the subsampled instances, while the mean of the anti-diagonal elements is assigned to the other subsampled instance. The specific description of a hyperspectral diagonal subsampler $S = (s_1, s_2)$ is outlined below.

1) The hyperspectral data $\mathbf{X}$ is partitioned into $[R/m] \times [C/m]$ cubes with each of shape $m \times m \times B$. In this article, m is set as 2.
2) The elements along the diagonal and anti-diagonal are chosen in the ith row and jth column cube at random. The subsampler is conducted using the convolution operator. More specifically, one element is derived through a 2-D convolution as

$$\mathbf{c_1} = \begin{bmatrix} 0 & 0.5 \\ 0.5 & 0 \end{bmatrix} \tag{1}$$

while the other remaining element is acquired by applying a 2-D convolution as

$$\mathbf{c_2} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}. \tag{2}$$

The operator is implemented executed band-wise rendering the diagonal subsampling strategy adaptive to any unspecified amount of input spectral bands. Consider them as the (i, j)th composition of the subsampler $S = (s_1, s_2)$.
3) Given each $[R/m] \times [C/m]$ cube, repeat step 2. For the hyperspectral data $\mathbf{X}$, paired subsampled hyperspectral instances $[s_1(\mathbf{X}), s_2(\mathbf{X})]$ are generated with shape $[R/m] \times [C/m] \times B$.

The above process allows us to employ various random diagonal averaging subsamplers to obtain paired hyperspectral instances from an original unlabeled HSI. The gap between the underlying backgrounds of paired HSIs $[s_1(\mathbf{X}), s_2(\mathbf{X})]$ is

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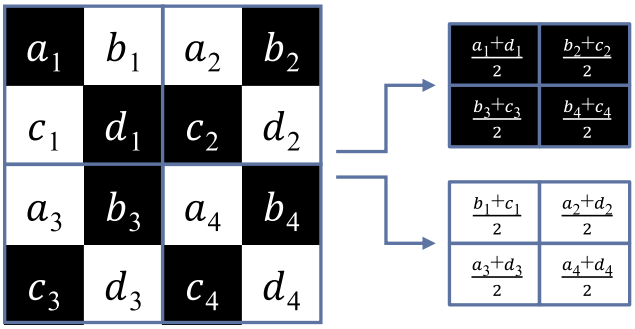


Fig. 4. Demonstration of deriving an image pair using a diagonal averaging subsampler $S = (s_1, s_2)$. In this regard, the elements along the diagonal and anti-diagonal are chosen in each 2×2 cube at random, occupied in black and white. The element occupied in black is considered as an element of a subsampled data $s_1(\mathbf{X})$, while the element occupied in white is considered as an element of another subsampled data $s_2(\mathbf{X})$. In the given scheme, there is one $4 \times 4 \times B$ data, generating paired instances of shape $2 \times 2 \times B$.

small because each spectral band on the identical location of $[s_1(\mathbf{X}), s_2(\mathbf{X})]$ in the source hyperspectral data is adjacent.

Analyzing from the perspective of image characteristics, the anomalous targets in hyperspectral data exhibit the succeeding features: 1) the characteristics of abnormal spectrum and non-anomalous background spectrum are significantly distinct; 2) the proportion of anomalous targets in the overall scenario is relatively small; and 3) the shape of anomalous targets shows a high degree of irregularity. Mathematically speaking, an original HSI $\mathbf{X}$ consisting of anomalous targets is usually modeled as

$$\mathbf{X} = \mathbf{Y} + \boldsymbol{\alpha}. \tag{3}$$

In the setting of embedding anomalous targets in the expression of irregular and nonuniform units, we assume that another authentic HSI of the same scene is represented as

$$\mathbf{Z} = \mathbf{Y} + \boldsymbol{\beta} \tag{4}$$

where $\mathbf{Y}$ represents the hyperspectral background (ground truth), $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ represent anomalous targets embedded in the background.

Given such an image pair $\{\mathbf{X}, \mathbf{Z}\}$, the background reconstruction neural network $\mathcal{F}_\theta$ can be trained by minimizing the squared-ℓ_2 loss in terms of θ

$$\arg\min_\theta \mathbb{E}_{\boldsymbol{\alpha},\boldsymbol{\beta}}\left\{\|\mathcal{F}_\theta(\mathbf{X}) - \mathbf{Z}\|_2^2\right\}. \tag{5}$$

The employed constraint bears close relevance to the one utilized in supervision learning

$$\arg\min_\theta \mathbb{E}_{\boldsymbol{\alpha}}\left\{\|\mathcal{F}_\theta(\mathbf{X}) - \mathbf{Y}\|_2^2\right\}. \tag{6}$$

We can prove the above conclusion through the following mathematical derivation process:

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{\alpha},\boldsymbol{\beta}}\left\{\|\mathcal{F}_\theta(\mathbf{X}) - \mathbf{Z}\|_2^2\right\} \\
&= \mathbb{E}_{\boldsymbol{\alpha},\boldsymbol{\beta}}\left\{\|\mathcal{F}_\theta(\mathbf{X}) - \mathbf{Y} - \boldsymbol{\beta}\|_2^2\right\} \\
&= \mathbb{E}_{\boldsymbol{\alpha},\boldsymbol{\beta}}\left\{\|\mathcal{F}_\theta(\mathbf{X}) - \mathbf{Y}\|_2^2 - 2(\boldsymbol{\beta})^\top(\mathcal{F}_{\boldsymbol{\theta}}(\mathbf{X}) - \mathbf{Y}) + (\boldsymbol{\beta})^\top\boldsymbol{\beta}\right\} \\
&= \mathbb{E}_{\boldsymbol{\alpha},\boldsymbol{\beta}}\left\{\|\mathcal{F}_\theta(\mathbf{X}) - \mathbf{Y}\|_2^2\right\} - 2\mathbb{E}_{\boldsymbol{\alpha},\boldsymbol{\beta}}\left\{(\boldsymbol{\beta})^\top\mathcal{F}_\theta(\mathbf{X})\right\} + \text{const.}
\end{aligned} \tag{7}$$

Clearly, given that the anomalous targets $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ remain independent of each other, it implies $\mathbb{E}_{\boldsymbol{\alpha},\boldsymbol{\beta}}\{(\boldsymbol{\beta})^\top\mathcal{F}_\theta(\mathbf{X})\} = 0$. The expectation of a constraint on $(\mathbf{X}, \mathbf{Z})$ will be comparable to the supervision learning constraint on $(\mathbf{X}, \mathbf{Y})$. However, in the real world, capturing multiple HSIs of the same scene containing anomalous targets is a highly challenging problem. We need to further explore new solutions in self-supervised learning.

Inspired by the above idea, the data pair subsampler $S = (s_1, s_2)$ is utilized to obtain a hyperspectral data pair $[s_1(\mathbf{X}), s_2(\mathbf{X})]$ from the original hyperspectral data $\mathbf{X}$. The information contained in paired instances $[s_1(\mathbf{X}), s_2(\mathbf{X})]$ are highly similar. Therefore, following (5), we attempt to consider the paired instances as observations of two different hyperspectral scenes containing anomalous targets, resulting in

$$\arg\min_\theta \mathbb{E}_{\boldsymbol{\alpha}}\left\{\|\mathcal{F}_\theta(s_1(\mathbf{X})) - s_2(\mathbf{X})\|_2^2\right\}. \tag{8}$$

In contrast to (5), there is a certain gap in the hyperspectral background (ground truth) of two sampled images $\{s_1(\mathbf{X}), s_2(\mathbf{X})\}$, i.e., the derivation process of (7) is not entirely applicable to some extent. Therefore, executing algorithm details in a similar manner based on the theory of (8) is inappropriate. We need to consider further optimizations.

Regarding a perfect background reconstruction neural network $\mathcal{F}_\theta{}^*$ under the optimal condition of supervised training, for an original HSI $\mathbf{X}$, it ensures these: $\mathcal{F}_\theta{}^*(\mathbf{X}) = \mathbf{Y}$, $\mathcal{F}_\theta{}^*(s_1(\mathbf{X})) = s_1(\mathbf{Y})$, and $\mathcal{F}_\theta{}^*(s_2(\mathbf{X})) = s_2(\mathbf{Y})$. Consequently, the subsequent statement remains true for the ideal model $\mathcal{F}_\theta{}^*$

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{\alpha}}\{\mathcal{F}_\theta{}^*(s_1(\mathbf{X})) - s_2(\mathbf{X}) - (s_1(\mathcal{F}_\theta{}^*(\mathbf{X})) - s_2(\mathcal{F}_\theta{}^*(\mathbf{X})))\} \\
&= s_1(\mathbf{Y}) - \mathbb{E}_{\boldsymbol{\alpha}}\{s_2(\mathbf{X})\} - (s_1(\mathbf{Y}) - s_2(\mathbf{Y})) \\
&= s_2(\mathbf{Y}) - \mathbb{E}_{\boldsymbol{\alpha}}\{s_2(\mathbf{X})\} = 0.
\end{aligned} \tag{9}$$

Equation (9) calculates the equality relationship satisfied when the background reconstruction neural network $\mathcal{F}$ is an ideal neural network $\mathcal{F}^*$. To achieve this ideal situation while ensuring consistency in the hyperspectral reconstruction, given the subsampled image pair $\{s_1(\mathbf{X}), s_2(\mathbf{X})\}$, we train the neural network $\mathcal{F}_\theta$ by reducing the following reconstruction consistency loss:

$$\mathcal{L} = \|\mathcal{F}_\theta(s_1(\mathbf{X})) - s_2(\mathbf{X}) - (s_1(\mathcal{F}_\theta(\mathbf{X})) - s_2(\mathcal{F}_\theta(\mathbf{X})))\|_2^2 \tag{10}$$

where $\mathcal{F}_\theta$ is a background reconstruction neural network designed in Section II-B. The complete running framework is depicted in Algorithm 1.

D. Detection Scheme

In the detection scheme, we feed the original HSI into a competently trained network $\mathcal{F}_{\hat{\theta}}$, characterized by the parameter $\hat{\theta}$. We obtain a hyperspectral background

$$\widehat{\mathbf{X}} = \mathcal{F}_{\hat{\theta}}(\mathbf{X}). \tag{11}$$

Following the self-supervised training process based on diagonal averaging subsampled images, the proposed DSRNet can be regarded as a reliable reconstructing model. Elements corresponding to anomalous targets in the source data are rebuilt to be nearly comparable to the adjacent background elements. Consequently, the output HSI can be regarded as

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a background for the original HSI scene, and anomalies can be regarded as crucial reconstructing errors in this process. Ultimately, we calculate the reconstructing errors to obtain the project outcome:

$$\mathbf{D}_{\mathbf{m},\mathbf{n}} = \|\mathbf{x}_{\mathbf{m},\mathbf{n}} - \widehat{\mathbf{x}}_{\mathbf{m},\mathbf{n}}\|_2 \tag{12}$$

where $\mathbf{x}_{\mathbf{m},\mathbf{n}} \in \mathbb{R}^{B\times 1}$ and $\widehat{\mathbf{x}}_{\mathbf{m},\mathbf{n}} \in \mathbb{R}^{B\times 1}$ denote the elements at (m, n) of $\mathbf{X}$ and $\widehat{\mathbf{X}}$. $\mathbf{D}_{\mathbf{m},\mathbf{n}}$ represents the possibility of anomaly at the respective element position, forming the eventual result map $\mathbf{D} \in \mathbb{R}^{R\times C}$.

Algorithm 1 DSRNet for Anomaly Detection

Input: Original HSI $\mathbf{X} \in \mathbb{R}^{R\times C\times B}$; Reconstruction network $\mathcal{F}_\theta$.
Result: Ultimate detection map $\mathbf{D} \in \mathbb{R}^{R\times C}$.

Stage 1: Training Scheme
while *not converged* **do**
  Gain a random diagonal subsampler $S = (s_1, s_2)$;
  Generate a pair of subsampled HSIs $(s_1(\mathbf{X}), s_2(\mathbf{X}))$, where $s_1(\mathbf{X})$ is the model input, and $s_2(\mathbf{X})$ is the optimization target of $s_1(\mathbf{X})$;
  For the model input $s_1(\mathbf{X})$, obtain the output data $\mathcal{F}_\theta(s_1(\mathbf{X}))$;
  For the original HSI $\mathbf{X}$, obtain the reconstructed HSI background $\mathcal{F}_\theta(\mathbf{X})$;
  Employ the identical diagonal subsampler S to obtain the pair $(s_1(\mathcal{F}_\theta(\mathbf{X})), s_2(\mathcal{F}_\theta(\mathbf{X})))$;
  Compute $\mathcal{L} = \|\mathcal{F}_\theta(s_1(\mathbf{X})) - s_2(\mathbf{X}) - (s_1(\mathcal{F}_\theta(\mathbf{X})) - s_2(\mathcal{F}_\theta(\mathbf{X})))\|_2^2$;
  Update the reconstruction model $\mathcal{F}_\theta$ by reducing the constraint function $\mathcal{L}$;
  Obtain the optimal model $\mathcal{F}$, parameterized by $\hat{\theta}$.
end
Stage 2: Detection Scheme
Gain the outcome hyperspectral background $\widehat{\mathbf{X}}$ by inputting the original HSI $\mathbf{X}$ to the competently trained reconstruction model $\mathcal{F}_{\hat{\theta}}$ by Eq. (11);
Compute the anomaly detection result $\mathbf{D}_{\mathbf{m},\mathbf{n}}$ for all element in the original HSI using Eq. (12).

III. Experiments and Analysis

In this section, to demonstrate the comprehensive detection performance of the proposed methodology, four publicly accessible benchmark hyperspectral data are implemented for comparative experiments, followed by a detailed analysis of the experimental outcomes.

A. Experimental Data

1) Beach Data: The first data was obtained from the Reflective Optics System Imaging Spectrometer (ROSIS-03) in a beach area of Pavia [57]. The anomalies in this dataset primarily include several vehicles on the bridge.

TABLE I
Detailed Information About the Experimental Data

Data	Image size	Location	Sensor	Resolution
Beach	150×150×102	Pavia	ROSIS-03	1.3m
HYDICE Urban	100×80×162	California	HYDICE	1m
Hyperion	150×150×155	Indiana	Hyperion	10m
Salinas	120×120×204	Salinas	AVIRIS	3.7m

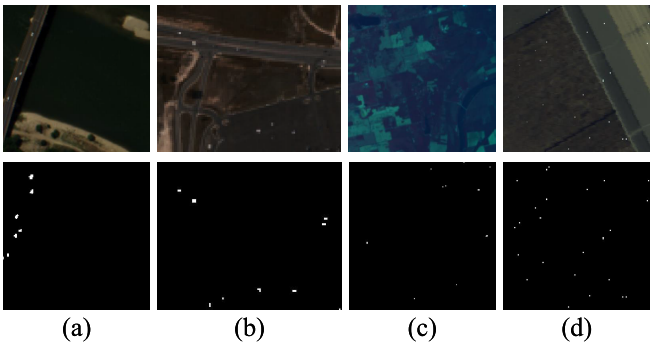


Fig. 5. Pseudocolor images and ground-truth maps of four experimental datasets. (a) Beach. (b) HYDICE Urban. (c) Hyperion. (d) Salinas.

2) HYDICE Urban Data: The second data was obtained from the Hyperspectral Data Image Collection Experiment (HYDICE) sensor in an urban area of California [24]. The anomalies in this dataset primarily include several vehicles and roofs.

3) Hyperion Data: The third data was obtained from the Earth Observing One (EO-1) Hyperion sensor in a farm area in Indiana [58]. The anomalies in this dataset mainly include storage boxes and roofs.

4) Salinas Data: The fourth data was obtained based on the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor from the Salinas Valley, CA, USA. The anomalies in this dataset mainly include storage boxes and roofs. The source dataset consists of information pertaining to uncovered ground, greeneries, and grape plantations, captured at a spatial resolution of 3.7 m. The target embedding approach [59] was employed to introduce anomalous targets into the AVIRIS Salinas sub-images, as described below

$$\mathbf{x} = \tau \times \mathbf{z} + (1 - \tau) \times \mathbf{y} \tag{13}$$

where $\mathbf{x}$ is the spectral vector of the synthesized subpixel anomalous target, $\mathbf{y}$ is the spectral vector of the background, $\mathbf{z}$ is the anomalous spectral vector, and τ is the specified abundance fraction with a range of 0.04–1. The details of the experimental data are listed in Table I and Fig. 5.

B. Experimental Settings

1) Comparison Methods: To assess the performance of the proposed DSRNet, several advanced detection algorithms are exploited for comparison, including LRX [12], CRD [32], SSFSCRD [33], VABS [39], CRDBPSW [34], GRX [10], FEBPAD [38], RGAE [51], GAED [55], and GNLTR [40].

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TABLE II
AUC SCORES OF (TPR, FPR) ON DIFFERENT DETECTORS FOR THE FOUR DATASETS

Scene	Detectors and AUC scores of (TPR, FPR)										
	GRX	LRX	CRD	SSFSCRD	FEBPAD	VABS	CRDBPSW	GNLTR	RGAE	GAED	DSRNet
Beach	0.9537	0.9198	0.9694	0.9830	0.9643	0.9449	0.9506	0.9748	0.9042	0.9374	**0.9928**
HYDICE Urban	0.9762	0.9824	0.9843	0.9087	0.9812	0.9123	0.9742	0.9829	0.8380	0.9937	**0.9984**
Hyperion	0.9896	0.9003	0.9742	0.9825	0.9832	0.9870	0.9943	0.9826	0.9367	0.9843	**0.9970**
Salinas	0.8073	0.9196	0.9437	0.9663	0.9783	0.9267	0.9908	0.9835	0.9286	0.9398	**0.9981**
Average	0.9317	0.9305	0.9679	0.9601	0.9768	0.9427	0.9775	0.9810	0.9019	0.9638	**0.9966**

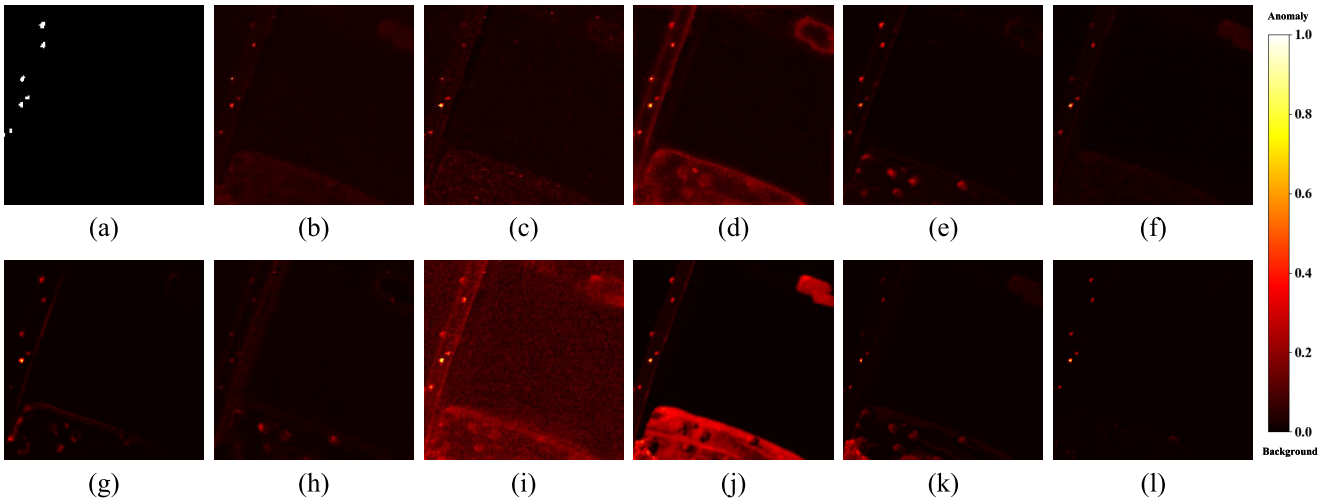


Fig. 6. Detection maps of different algorithms for the Beach scene. (a) Ground truth. (b) GRX. (c) LRX. (d) CRD. (e) SSFSCRD. (f) FEBPAD. (g) VABS. (h) CRDBPSW. (i) GNLTR. (j) RGAE. (k) GAED. (l) DSRNet.

Among them, the RGAE and GAED algorithms adopted deep learning structures, aiming to enhance the ability of the model to rebuild the background. To quantify the anomaly degree of individual elements in the scene, residue signals derived from the disparity between the original hyperspectral data and its rebuilt counterpart are utilized. This idea aligns with the proposed DSRNet and provides a theoretical foundation. For the proposed DSRNet, we employ the Adam optimizer for all the datasets, setting the learning rate to 0.0001 and the weight decay to 0. All experiments were conducted employing using PyTorch 1.11.0 on a GPU of NVIDIA GeForce RTX 2080 Ti with 11 GB memory. For other non-deep learning anomaly detection algorithms, we keep the parameters consistent with those recommended in the original literature. For the GAED approach, the hidden layer comprises 25 nodes and the learning rate sets as 0.1. For the RGAE approach, we randomly initialize weights and biases with a superpixel number of 150, a tradeoff parameter of 0.01, and a hidden layer dimension of 100 in all experiments.

2) Evaluation Metrics: To measure the real detection performance of various algorithms, typically used assessment criteria include the receiver operating characteristic curve (ROC) [60], the resulting area under the ROC curve (AUC) [61], and the statistical separability boxplot.

The ROC curve represents the correlation between anomaly possibility and mistaken alert, providing a comprehensive detection assessment of the approach. As a widely adopted evaluation criterion in hyperspectral anomaly detection [62], the ROC curve is constructed based on three key parameters: true positive rate (TPR), false positive rate (FPR), and the threshold for the detection probability map (τ). The ROC curve effectively illustrates the transformation relationship between TPR and FPR, which can be derived through

$$\mathrm{TPR} = \frac{\mathrm{TP}}{N_a}, \quad \mathrm{FPR} = \frac{\mathrm{FP}}{N_b} \tag{14}$$

where TP denotes the count of accurately identified anomalies, while FP signifies the count of background elements incorrectly classified as anomalies, both based on detection probabilities surpassing a specified threshold τ. N_a and N_b represent the overall quantity of anomalies and background elements, respectively. An ideal algorithm characterized by significant TPR and insignificant FPR is represented by a ROC curve positioned close to the upper left corner.

The corresponding $\mathrm{AUC}_{(\mathrm{TPR,FPR})}$ value is used to quantitatively evaluate the detection result, which can be derived through

$$\mathrm{AUC}_{(\mathrm{TPR,FPR})} = \int_{-\infty}^{+\infty} \mathrm{TPR}(\tau) d\mathrm{FPR}(\tau). \tag{15}$$

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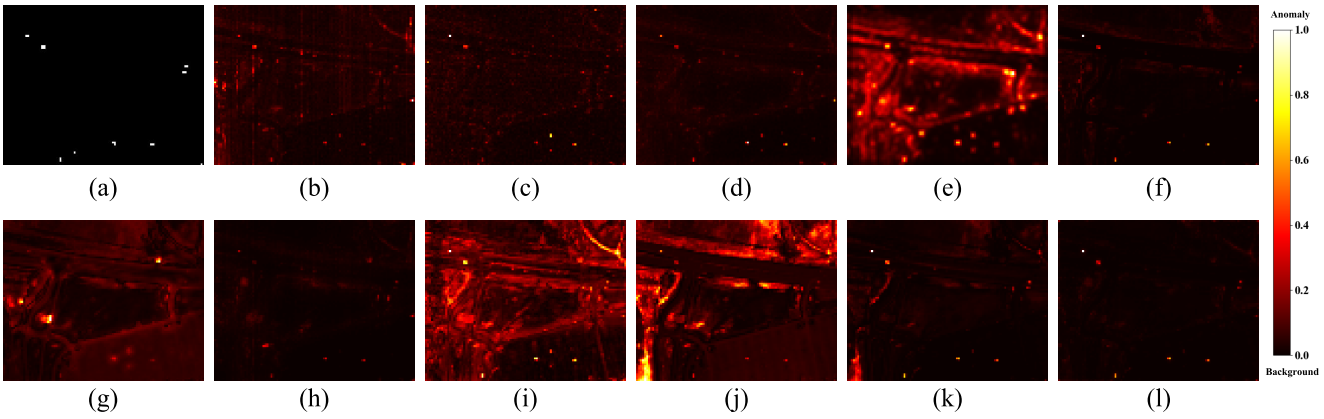


Fig. 7. Detection maps of different algorithms for the HYDICE Urban scene. (a) Ground truth. (b) GRX. (c) LRX. (d) CRD. (e) SSFSCRD. (f) FEBPAD. (g) VABS. (h) CRDBPSW. (i) GNLTR. (j) RGAE. (k) GAED. (l) DSRNet.

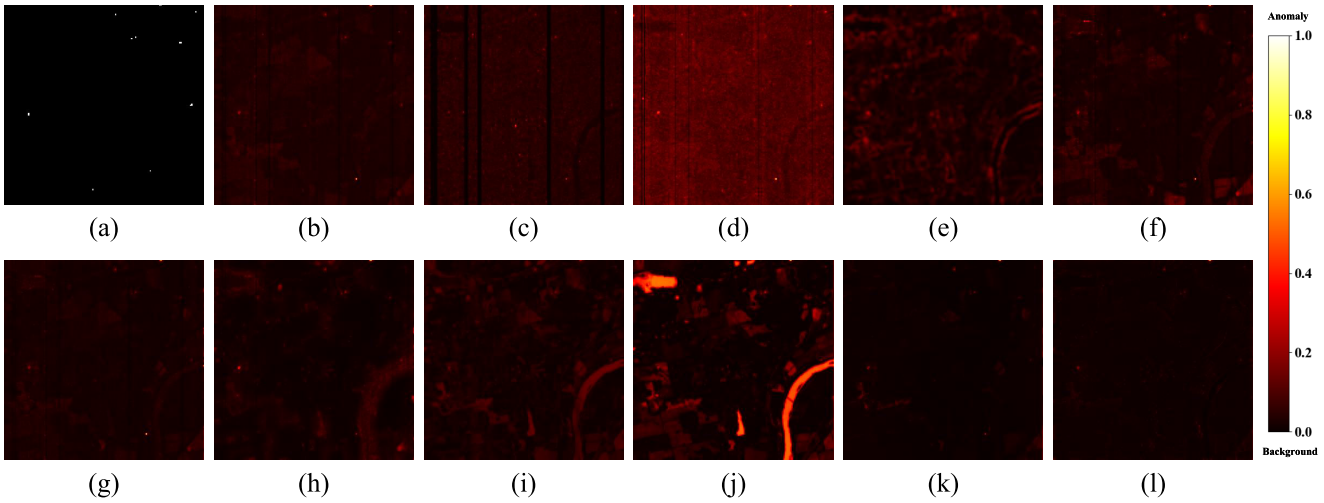


Fig. 8. Detection maps of different algorithms for the Hyperion scene. (a) Ground truth. (b) GRX. (c) LRX. (d) CRD. (e) SSFSCRD. (f) FEBPAD. (g) VABS. (h) CRDBPSW. (i) GNLTR. (j) RGAE. (k) GAED. (l) DSRNet.

According to [63], in addition to $\mathrm{AUC}_{(\mathrm{TPR},\mathrm{FPR})}$, another AUC indicator namely $\mathrm{AUC}_{(\mathrm{FPR},\tau)}$ is computed as the following expressions:

$$\mathrm{AUC}_{(\mathrm{FPR},\tau)} = \int_{-\infty}^{+\infty} \mathrm{FPR}(\tau) d\tau. \tag{16}$$

Greater values of $\mathrm{AUC}_{(\mathrm{TPR},\mathrm{FPR})}$ correspond to increased detection accuracy, whereas smaller values of $\mathrm{AUC}_{(\mathrm{FPR},\tau)}$ correlate with decreased false alarm rates. An ideal algorithm should have the ROC curve of (FPR, τ) close to the lower left corner, accompanied by the $\mathrm{AUC}_{(\mathrm{TPR},\mathrm{FPR})}$ value approaching 1.

Furthermore, the ability of the algorithm to distinguish anomalies from the background and suppress background elements can be appropriately assessed by the separability boxplot. The detection scores of all pixels in the scene are presented in distinct boxes, where the background category is represented by a cyan box and the anomaly category is represented by a red box. Each box spans from the 10th to the 90th percentile. Notably, a detector demonstrating superior capability in highlighting anomalies and suppressing backgrounds would have the red box positioned at the top, with a substantial gap between the red and the cyan box.

C. Experimental Results and Analysis

In this part, the efficacy of the proposed DSRNet is verified, and a comprehensive comparison of the detection results on various algorithms is conducted. Table II presents the $\mathrm{AUC}_{(\mathrm{TPR},\mathrm{FPR})}$ scores for various datasets, providing a numerical metric of the detection performance. The best detection performance is highlighted in bold, and the same applies throughout the rest of this article. Figs. 6–9 depict the detection results of different anomaly detection algorithms on four benchmark datasets. Furthermore, Figs. 10–13 display the ROC curves and statistical separability boxplots of these detectors, whose $\mathrm{AUC}_{(\mathrm{TPR},\tau)}$ scores are listed in Table III.

For Beach scene, the map of ground truth is depicted in Fig. 6(a), while the detection maps of various methods are presented in Fig. 6(b)–(l). The proposed DSRNet can effectively detect all vehicle targets on the bridge, dampening the anomaly probabilities of the background elements to a mild degree and displaying notably intense visual difference. GRX, LRX, SSFSCRD, FEBPAD, VABS, CRDBPSW, and GAED are able to detect some positions of anomalous targets while they yield a relatively high false alert rate at the bottom of the scene, failing to effectively highlight the anomalous targets

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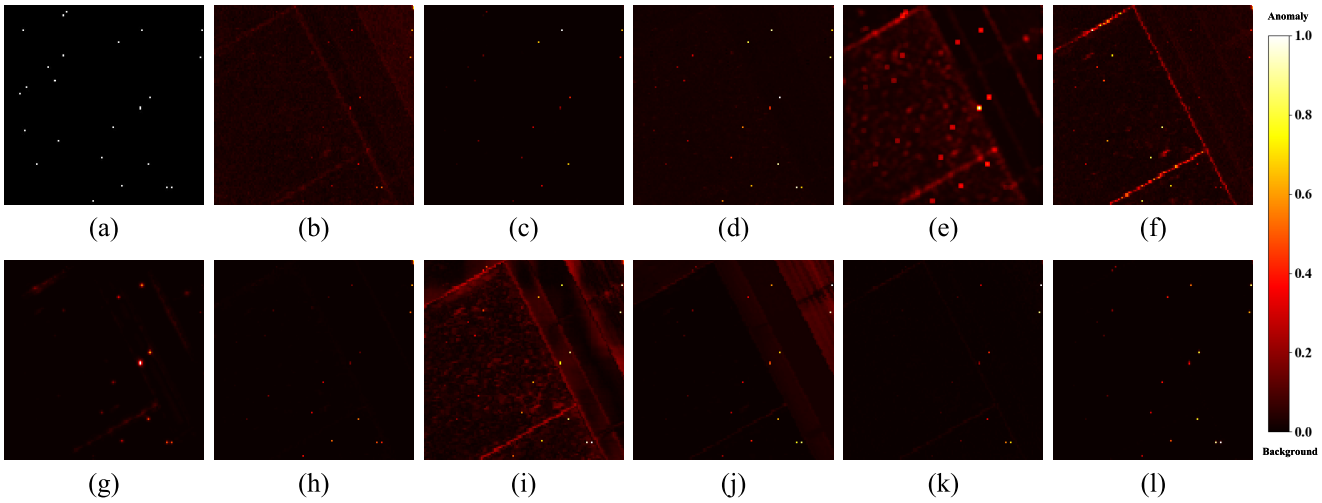
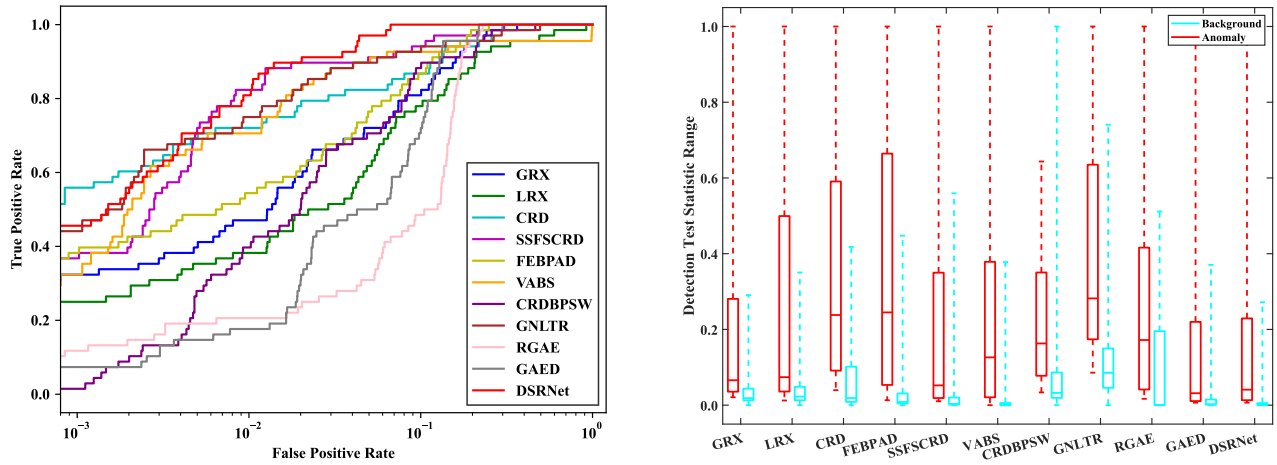


Fig. 9. Detection maps of different algorithms for the Salinas scene. (a) Ground truth. (b) GRX. (c) LRX. (d) CRD. (e) SSFSCRD. (f) FEBPAD. (g) VABS. (h) CRDBPSW. (i) GNLTR. (j) RGAE. (k) GAED. (l) DSRNet.



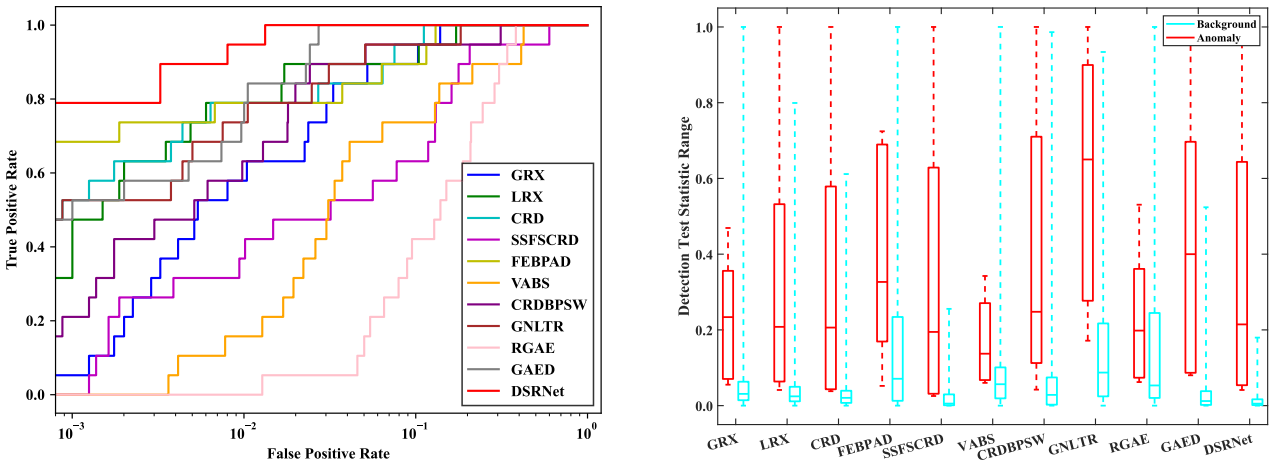
(a) (b)

Fig. 10. ROC curves and separability boxplot on Beach scene. (a) ROC curves. (b) Separability boxplot.

on the bridge. While CRD, GNLTR, and RGAE exhibit the ability to identify certain anomalous targets in the scene, they concurrently keep a significant amount of background structural knowledge, leading to an increased incidence of false alerts. The ROC curves and statistical separability boxplot in Fig. 10 substantiate the aforementioned inferences. The ROC curves for both SSFSCRD and our proposed DSRNet exhibit proximity to the upper left corner, signifying excellent performance in identifying anomalies. Furthermore, the TPR of DSRNet is optimal across a broad range of FPRs, particularly at approximately 10^{-1}. The statistical separability boxplot distinctly reveals that DSRNet exhibits least cyan box positioned exceedingly close to zero, which indicates its excellent capability in effectively dampening background interference. The AUC score attained by our proposed DSRNet is the highest at 0.9928, while maintaining the lowest false alarm rate.

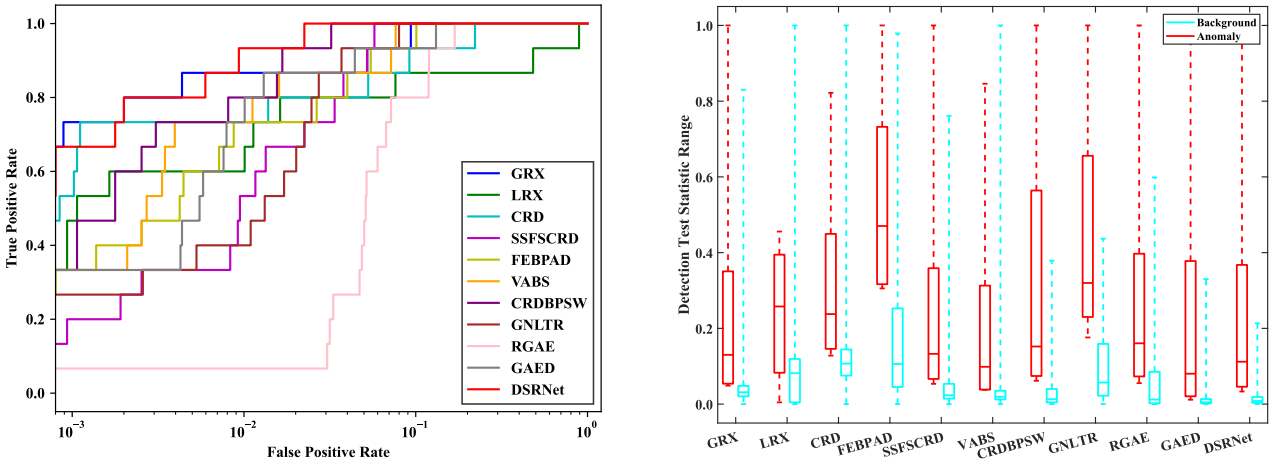
For HYDICE Urban scene, the map of ground truth is depicted in Fig. 7(a), while the maps of various methods are illustrated in Fig. 7(b)–(l). The proposed DSRNet preserves the complete form of all the vehicle and roof anomalous targets with relatively high certainty, and at the same time, can suppress the anomaly probabilities of background elements to a mild degree, displaying notably intense visual difference. CRD, FEBPAD, VABS, CRDBPSW, and GAED are able to detect some anomalous targets while they have difficulty distinguishing between anomalous targets and background elements. GRX, SSFSCRD, GNLTR, and RGAE collectively exhibit a tendency to incorrectly detect a large amount of background information, leading to a decrease in overall performance. The ROC curves and statistical separability boxplot in Fig. 11 substantiate the aforementioned inferences. The ROC curve derived by DSRNet exhibits the closest proximity to the upper left corner, symbolizing excellent results in identifying anomalies. In addition, the detection accuracy of DSRNet is optimal over a large range of FPRs, particularly at approximately 10^{-2}. In the statistical separability boxplot, it is evident that the red and cyan boxes of DSRNet are

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(a) (b)

Fig. 11. ROC curves and separability boxplot on HYDICE Urban scene. (a) ROC curves. (b) Separability boxplot.



(a) (b)

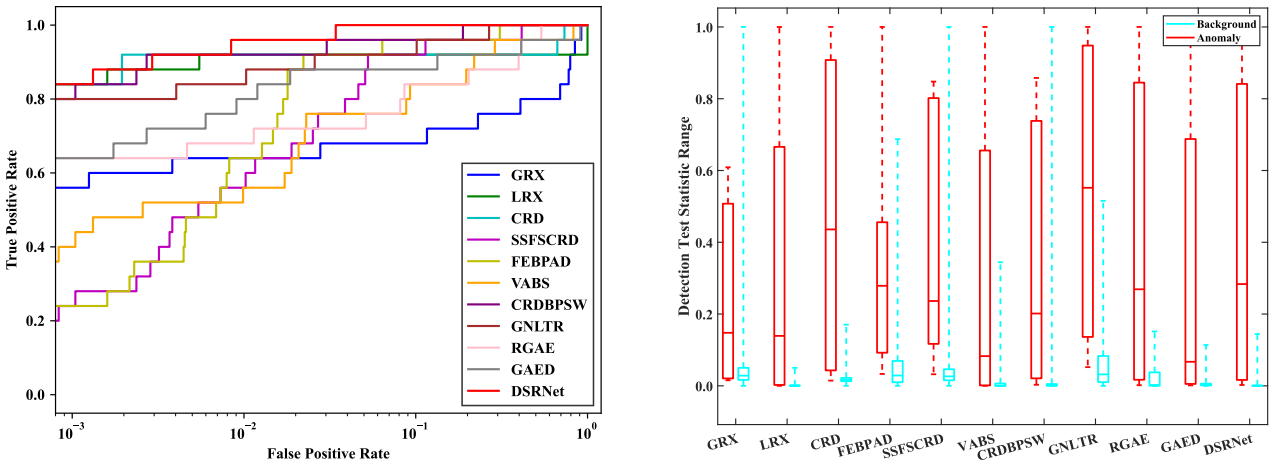
Fig. 12. ROC curves and separability boxplot on Hyperion scene. (a) ROC curves. (b) Separability boxplot.

absolutely isolated, with little overlap, signifying a superior distinction between the anomalous targets and the background elements. The AUC score attained by our proposed DSRNet is the highest at 0.9984, while maintaining the lowest false alarm rate.

For Hyperion scene, the map of ground truth is depicted in Fig. 8(a), while the detection maps of various algorithms are presented in Fig. 8(b)–(l). The proposed DSRNet can effectively detect all the storage box and roof anomalous targets, dampening the anomaly probabilities of the background elements to a mild degree and displaying notably intense visual difference. GRX, VABS, and CRDBPSW are able to identify the anomalous targets to a certain extent, but they incorrectly highlight background elements with large confidence, leading to a decrease in overall performance. LRX, CRD, GNLTR, and RGAE generally have a higher detection probability for abnormal targets, yet they exhibit strong certainty in incorrectly identifying edge regions, resulting in reduced detection accuracy. The ROC curves and statistical separability boxplot in Fig. 12 demonstrate the aforementioned inferences. The ROC curves derived by all other algorithms exhibit proximity to the upper left corner with the exception of RGAE. Additionally, the ROC curve for the proposed DSRNet is depicted by the red curve, illustrating remarkably identifying performance across various FPRs. The statistical separability boxplot clearly shows that the red and cyan boxes of DSRNet are almost absolutely separated, which signifies the remarkable results in background element suppression and anomalous target location. The AUC score attained by our proposed DSRNet is the highest at 0.9970, surpassing the second-highest CRDBPSW by 0.0027, which records a score of 0.9943.

For Salinas scene, the map of ground truth is depicted in Fig. 9(a), while the detection maps of various algorithms are illustrated in Fig. 9(b)–(l). DSRNet can effectively enhance the ability to locate anomalous targets while keeping background inhibition at a lower degree, resulting in detection outcomes that closely align with the ground truth. For the GRX and

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(a) (b)

Fig. 13. ROC curves and separability boxplot on Salinas scene. (a) ROC curves. (b) Separability boxplot.

TABLE III
AUC SCORES OF (FPR, τ) ON DIFFERENT DETECTORS FOR THE FOUR DATASETS

Scene	Detectors and AUC scores of (FPR, τ)										
	GRX	LRX	CRD	SSFSCRD	FEBPAD	VABS	CRDBPSW	GNLTR	RGAE	GAED	DSRNet
Beach	0.0233	0.0277	0.0383	0.0155	0.0079	0.0033	0.0452	0.0927	0.0376	0.0081	**0.0023**
HYDICE Urban	0.0380	0.0291	0.0229	0.1016	0.0116	0.0601	0.0370	0.1083	0.0984	0.0192	**0.0078**
Hyperion	0.0331	0.0788	0.1085	0.1326	0.0297	0.0222	0.0187	0.0771	0.0423	**0.0075**	0.0102
Salinas	0.0313	0.0007	0.0171	0.0374	0.0331	0.0026	0.0023	0.0417	0.0156	0.0034	**0.0006**
Average	0.0314	0.0341	0.0467	0.0718	0.0206	0.0221	0.0258	0.0800	0.0485	0.0096	**0.0052**

RGAE algorithms, due to a relatively high false alert rate, some anomalous edge details are mixed with the background elements. Although SSFSCRD, FEBPAD, and GNLTR exhibit the ability to identify certain anomalous targets in the scene, they concurrently preserve a substantial amount of background structural knowledge, resulting in an increased incidence of false alerts. Fig. 13 illustrates the ROC curves and the statistical separability boxplot. The ROC curve derived by our proposed DSRNet exhibits the closest proximity to the upper left corner, symbolizing excellent results in identifying anomalies. Furthermore, the TPR demonstrated by DSRNet is optimal across most FPRs, particularly at approximately 10^{-1}. The statistical separability boxplot distinctly reveals that the cyan and red boxes of DSRNet are almost absolutely isolated, with a notably significant gap, signifying a great distinction between the anomalous targets and the background elements. The AUC score attained by our proposed DSRNet is the highest at 0.9981, while maintaining the lowest false alarm rate.

D. Ablation Study

In this section, we evaluate the effectiveness of the proposed diagonal averaging subsampling strategy and reconstruction consistency constraint in the training scheme. We adopt a reconstruction model based on optimizing (8) as the BaseNet,

TABLE IV
AUC SCORES OF ABLATION STUDY FOR DIFFERENT DATASETS

Scene	Detectors and AUC scores		
	BaseNet	DSRNet without DS	DSRNet
Beach	0.9685	0.9812	**0.9928**
HYDICE Urban	0.9713	0.9830	**0.9984**
Hyperion	0.9727	0.9845	**0.9970**
Salinas	0.9802	0.9879	**0.9981**

which employs the same network and training process as the proposed DSRNet in this work, with the exception of replacing the diagonal averaging subsampling strategy by random subsampling in each data cube. The proposed DSRNet has innovatively introduced the reconstruction consistency constraint through comprehensive theoretical derivation, a principle that is reflected in the construction of a hyperspectral reconstruction framework utilizing the self-supervised training process. Here, we represent it as DSRNet without DS. At the same time, the diagonal averaging subsampler is introduced to generate paired training data, further purifying the background of the reconstructed scene and enhancing the anomaly detection capability of the detector. The comparison

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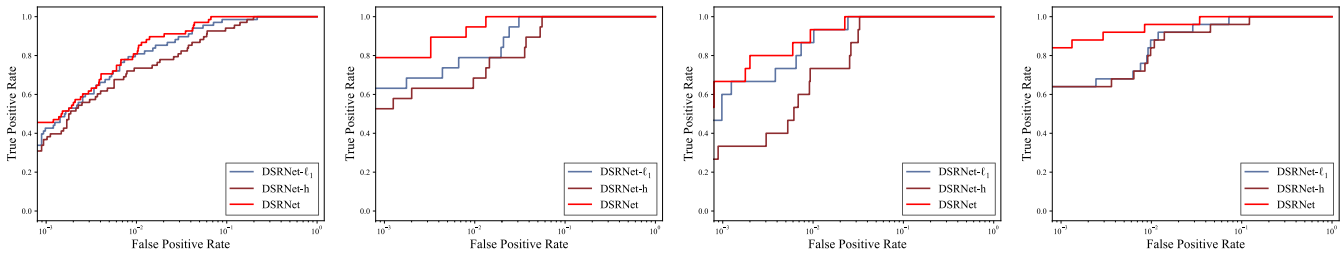


Fig. 14. ROC curves of (TPR, FPR) using different constraint functions on the four test data. (a) Beach. (b) HYDICE Urban. (c) Hyperion. (d) Salinas.

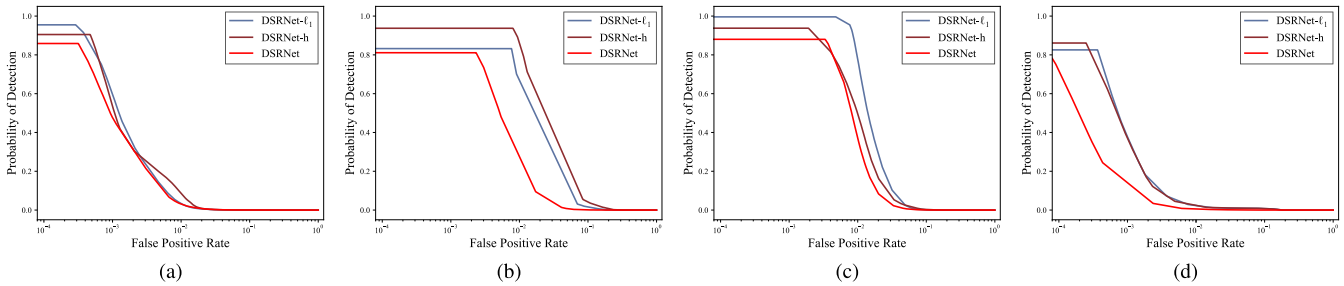


Fig. 15. ROC curves of (FPR, τ) using different constraint functions on the four test data. (a) Beach. (b) HYDICE Urban. (c) Hyperion. (d) Salinas.

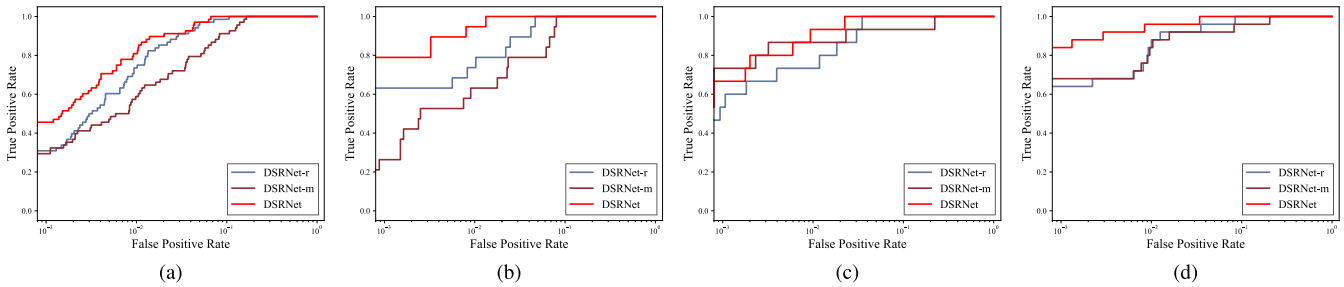


Fig. 16. ROC curves of (TPR, FPR) using different subsampling strategies on the four test data. (a) Beach. (b) HYDICE Urban. (c) Hyperion. (d) Salinas.

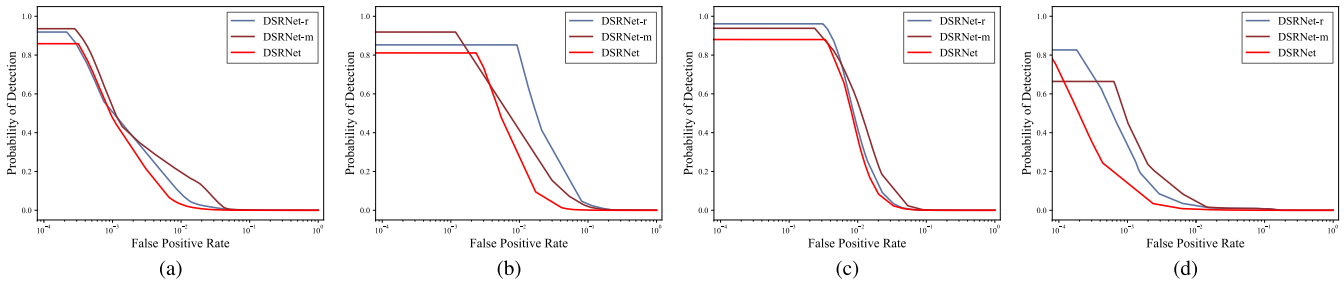


Fig. 17. ROC curves of (FPR, τ) using different subsampling strategies on the four test data. (a) Beach. (b) HYDICE Urban. (c) Hyperion. (d) Salinas.

of AUC scores on these models across four publicly available datasets is detailed in Table IV.

As observed from Table IV, the unsupervision-based DSRNet without using the diagonal averaging subsampling strategy has certain advantages compared with the BaseNet model, which is only well trained with a reconstruction consistency constraint, especially in the Salinas scene. Nevertheless, its performance improvement is still relatively limited due to the lack of effective training pair-generation strategy. On the contrary, when reconstructing the elements with the diagonal-averaging subsampling training pair, the ability of DSRNet to locate anomalous targets from hyperspectral scenes is markedly improved, resulting in higher AUC scores.

E. Model Analysis

1) Influence of Reconstruction Constraint: To validate the efficacy of the loss function in modeling the background reconstruction neural network, we substitute it from (10) using the ℓ_1 constraint. The DSRNet utilizing the ℓ_1 constraint is represented by DSRNet-ℓ_1. Furthermore, to further analyze the proposed framework, we conducted anomaly detection by

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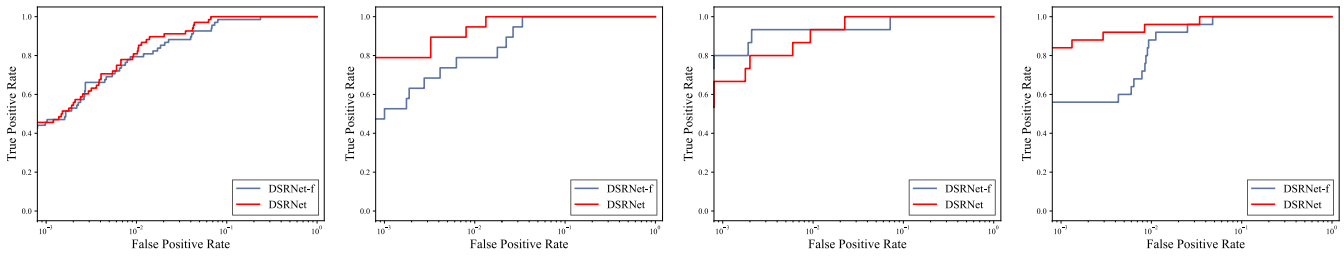


Fig. 18. ROC curves of (TPR, FPR) using fix-location sampling on the four test data. (a) Beach. (b) HYDICE Urban. (c) Hyperion. (d) Salinas.

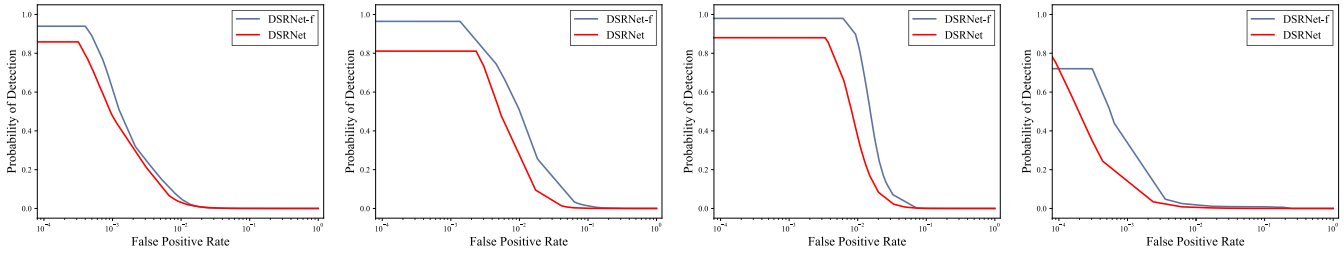


Fig. 19. ROC curves of (FPR, τ) using fix-location sampling on the four test data. (a) Beach. (b) HYDICE Urban. (c) Hyperion. (d) Salinas.

TABLE V
TIME CONSUMPTION (IN SECONDS) OF VARIOUS ALGORITHMS ON THE FOUR DATASETS

Data	GRX	LRX	CRD	SSFSCRD	FEBPAD	VABS	CRDBPSW	GNLTR	RGAE	GAED	DSRNet
Beach	0.3049	16.4432	2.8663	14.9446	5.4191	4.7487	33.2296	8.9541	202.0410	82.9222	39.3348
HYDICE Urban	0.2497	13.7913	2.3695	6.5507	2.9059	2.5427	17.4231	7.2613	56.0819	35.1768	29.5946
Hyperion	0.6598	22.8553	3.6128	7.9753	5.9049	4.9580	48.2589	9.7573	245.3744	97.1442	48.3763
Salinas	0.2933	21.8180	3.4632	11.5186	4.2629	3.5012	29.8189	11.1790	149.5405	58.6162	35.7985

optimizing (8), and named this model as DSRNet-h. To ensure an equitable comparison, all training parameter settings in DSRNet-ℓ_1, DSRNet-h, and DSRNet remain entirely consistent. The ROC curves for (TPR, FPR) and (FPR, τ) are depicted in Figs. 14 and 15, respectively. From Fig. 14, it becomes evident that, considering optimization schemes modeled with different mathematical principles, DSRNet-ℓ_1 and DSRNet-h with various false alert rates have their own superiorities. Nevertheless, upon combining the reconstruction constraint from both two perspectives, namely DSRNet, there is an enhancement in detection results to different extents, particularly for the HYDICE Urban dataset and Salinas dataset. The similar inference could be drawn from Fig. 15, indicating the superior capability of DSRNet for suppressing background interference relative to DSRNet-ℓ_1 and DSRNet-h.

2) Influence of Subsampling Strategy: To explore the influence of subsampling strategy on the detection performance of DSRNet, we evaluated the proposed framework using random average subsampling and diagonal maximization subsampling. The random average subsampling strategy involves the random sampling of each data cube in the original HSI, where two elements are randomly selected and averaged to assign to one subsampled image, and the remaining two elements are averaged and assigned to another subsampled image. The diagonal maximization subsampling strategy is a variation of diagonal averaging subsampling. It involves subsampling each data cube in the original HSI, extracting the maximum value of elements on the diagonal, then randomly assigning it to one subsampled image. At the same time, the maximum value of elements on the anti-diagonal is extracted and randomly assigned to the remaining other subsampled image. The DSRNet base on the random average subsampling strategy is denoted as DSRNet-r, and the one base on the diagonal maximization subsampling is denoted as DSRNet-m. The ROC curves for (TPR, FPR) and (FPR, τ) are provided in Figs. 16 and 17, respectively. From Fig. 16, the identifying performance of DSRNet-r and DSRNet-m over various false alert rates exhibits unique advantages. Nevertheless, when we adopt the diagonal averaging subsampling strategy proposed in Section II-C, the identifying performance has been boosted to different extents, particularly for the Salinas dataset. We could draw a consistent inference from Fig. 17, showing that DSRNet has a stronger capability to suppress background information compared to DSRNet-r and DSRNet-m.

3) Influence of Fix-Location Sampling or Not: Fix-location sampling involves assigning the average values of the diagonal elements for each cube in the original HSI to subsampled image I, and the average values of the anti-diagonal elements

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for each cube to image II. Compared to anomaly detection models using a fix-location sampling form, detection models employing the non-fix-location sampling are more informative intuitively. To evaluate the effect of this variation in identifying model, we substitute it from Section II-C using the fix-location sampling. The DSRNet with fix-location sampling is named DSRNet-f. To ensure an equitable comparison, all training parameter settings in DSRNet and DSRNet-f remain entirely consistent. The ROC curves of (TPR, FPR) and (FPR, τ) derived by the DSRNet-f and DSRNet are depicted in Figs. 18 and 19, respectively. From Fig. 18, we can observe that for the Beach and Hyperion datasets, DSRNet and DSRNet-f show unique advantages under various false alert rates. For the HYDICE Urban and Salinas datasets, the anomaly detection performance of DSRNet is superior to DSRNet-f. The similar inference is able to be drawn from Fig. 19. In general, DSRNet exhibits a better capability of background information suppression compared to DSRNet-f.

F. Computing Costs

The time consumptions of various algorithms on all test datasets are presented in Table V to assess the computational costs of these models. The software environment of all algorithms is built with CUDA 11.7, Python 3.8.18, and PyTorch 1.11.0 for the experiments. Due to its clean and effective architecture design, the computing cost of DSRNet is excellently controlled, which has a lower cost than other deep learning methods such as RGAE and GAED. The clean architecture design and efficient detection performance demonstrate its potential applications in real-world contexts. Furthermore, compared to other statistical and representation algorithms, GRX exhibits the least time consumption with its simplest design principles.

IV. Conclusion

In this article, a novel framework named DSRNet is presented to train an effective anomaly detection model using only the original hyperspectral data without any prior knowledge. More specifically, the framework contains a paired training data generation algorithm and a self-supervision training process enforced with a reconstruction consistency constraint. DSRNet first develops a randomly diagonal averaging subsampler to obtain training pairs from the same hyperspectral data and the elements corresponding to the two instances are cross-adjacent in the original data. On the other hand, a simple and reliable optimization function is exploited for ensuring the consistency of hyperspectral reconstruction in the self-supervised training scheme by rigorous mathematical proofs. Extensive experiments demonstrate that DSRNet has superior performance in anomalous target detection over four publicly benchmarks (Beach, HYDICE Urban, Hyperion, Salinas), with an average AUC score increase of 0.0156 and excellent suppression capability on background information.

In our future work, we will further integrate the statistical theory into anomaly detection algorithms on hyperspectral data to enhance their robustness and verify their effectiveness in potential applications.

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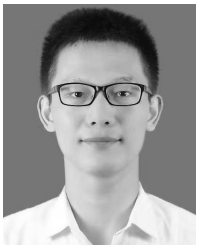
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